

2.2: Point Groups

The Low Symmetry Point Groups

C_1 Point Group

Overall, we divide point groups into three major categories: High symmetry point groups, low symmetry point groups, dihedral point groups, and rotational point groups. Let us begin with the low symmetry point groups. As the name says, these point groups only have few symmetry elements and operations. The point group C_1 is the point group with the lowest symmetry. Molecules that belong to this point group only have the identity as symmetry element.

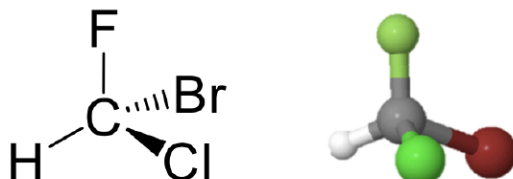


Figure 2.2.1 C_1 point group of bromochlorofluoromethane (Attribution: symotter.org/gallery)

An example is the bromochlorofluoromethane molecule (Fig. 2.2.1). It has no symmetry element, but the identity. The name C_1 comes from the symmetry element C_1 . A C_1 operation is the same as the identity.

C_s Point Group

The point group C_s has a mirror plane in addition to the identity. An example is the 1,2-bromochloroethene molecule (Fig. 2.2.2).

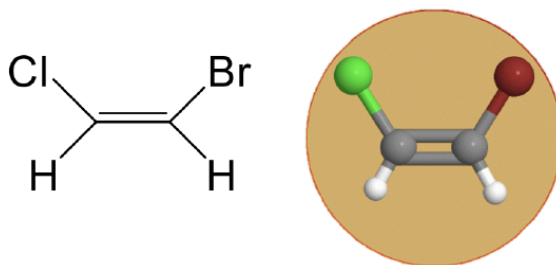


Figure 2.2.2 C_s point group of 1,2-bromochloroethene
(Attribution: symotter.org/gallery)

This is a planar molecule and the mirror plane is within the plane of the molecule. This mirror plane does not move any atoms when the reflection operation is carried out, nonetheless it exists because any point of the molecule above the mirror plane will be found below the mirror plane after the execution of the operation. Vice versa, any point below the mirror plane will be above the mirror plane. This mirror plane does not have a vertical or horizontal mirror plane designation because no proper rotational axes exist.

C_i Point Group

The point group C_i has the inversion as the only symmetry element besides the identity. The point group C_i is sometimes also called S_2 because an S_2 improper rotation-reflection is the same as an inversion. An example is the 1,2-dibromo 1,2-dichloro ethane (Fig. 2.2.3).

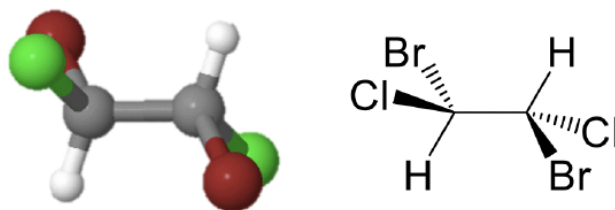


Figure 2.2.3 The C_i point group: 1,2-dibromo 1,2-dichloro ethane (Attribution: symotter.org/gallery)

This molecule looks quite symmetric, but it has inversion center in the middle of the carbon-carbon bond as the only symmetry element. Upon execution of the inversion operation, the two carbons swap up their positions, and so do the two bromine, the two chlorine, and the two hydrogen atoms.

The High Symmetry Point Groups

The symmetry elements of the high symmetry point groups can be more easily understood when the properties of platonic solids are understood first. Platonic solids are polyhedra made of regular polygons. In a platonic solid all faces, edges, and vertices (corners) are symmetry-equivalent. We will see that this is a property that can be used to understand the symmetry elements in high symmetry point groups. There are only five possibilities to make platonic solids from regular polygons (Fig. 2.2.4).

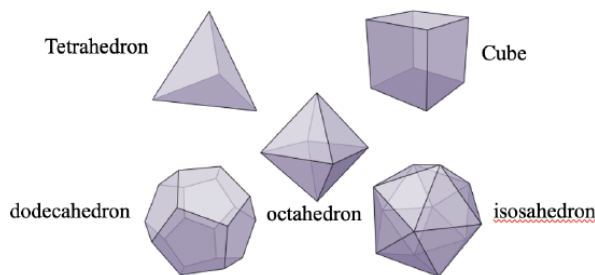


Figure 2.2.4 The platonic solids (Attribution: Drummyfish [CC0] <https://commons.wikimedia.org/wiki/File:Transparent.svg>)

The first possibility is to construct a tetrahedron from four regular triangles. The second platonic solid is the octahedron made of eight regular triangles. The third possibility is the icosahedron made of twenty triangles. In addition, six squares can be connected to form a cube, and twelve pentagons can be connected to form a dodecahedron. There are no possibilities to connect other regular polygons like hexagons to make a platonic solid.

The icosahedron is the most complex of all platonic solids. If you would like to see and study an icosahedron from the outside and inside, there is one for study on the playground of the Allentown Cedar Beach Park, in Allentown, Pennsylvania.



Figure 2.2.5 Icosahedron at Allentown Cedar Beach Park

The T_d Point Group

The tetrahedron, as well as tetrahedral molecules and anions such as CH_4 and BF_4^- belong to the high symmetry point group T_d . Let us find the symmetry elements and symmetry operations that belong to the point group T_d . First, we should not forget the identity operation, E . Next, it is useful to look for the principal axes.

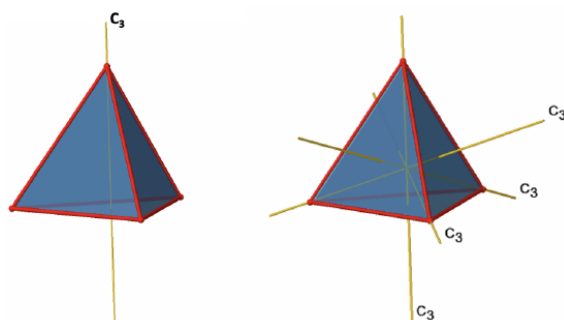


Figure 2.2.6 The C_3 axes in a tetrahedron (Attribution: symotter.org/gallery)

The tetrahedron has four principal C_3 axes (Fig. 2.2.6). It is a property of the high-symmetry point groups that they have more than one principal axis. The C_3 axes go through the vertices of the tetrahedron. Because each C_3 axis goes through one vertex, there are four vertices, and we know that in a platonic solid all vertices are symmetry-equivalent, we can understand that there are four C_3 axes. How many unique C_3 operations are associated with these axes? After three rotations around 120° we reach the identity. Therefore $C_3^3=E$, and we only need to consider the C_3^1 and the C_3^2 rotation about 120 and 240° respectively. Because there are four C_3 axes, there are four C_3^1 and four C_3^2 operations and eight C_3 operations overall. We can express this by writing the respective numbers as coefficients in front of the Schoenflies symbol for the operations (Fig. 2.2.7).

Symmetry operations: $E, 8C_3$

Symmetry operations in detail: $E, 4 C_3^1, 4 C_3^2$ ($C_3^3 = E$)

Figure 2.2.7 Symmetry operations associated with the C_3 axes in the point group T_d

In addition to the C_3 axes there are C_2 axes (Fig. 2.2.8).

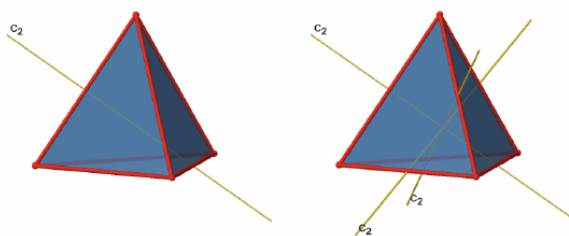


Figure 2.2.8 The C_2 axes in a tetrahedron belonging to the point group T_d (Attribution: symotter.org/gallery)

You can see that a C_2 axis goes through two opposite edges in the tetrahedron. Because a tetrahedron has six edges, and each C_2 axis go through two edges there are $6/2=3$ C_2 axes. There is only one C_2 symmetry operation per C_2 axis because we produce the identity already after two rotations. Therefore there are three C_2^1 operations overall (Fig. 2.2.9).

Symmetry operations: $3C_2$

Symmetry operations in detail: $3 C_2^1$

Figure 2.2.9 Symmetry operations associated with the C_2 axes in the point group T_d

In addition, the T_d point group has S_4 improper rotation reflections. Like the C_2 axes, they pass through the middle of two opposite edges. This also means that they are superimposing the C_2 axes. Because there are six edges, and two S_4 axes per edge there are $6/2=3$ S_4 axes (Fig. 2.2.10).

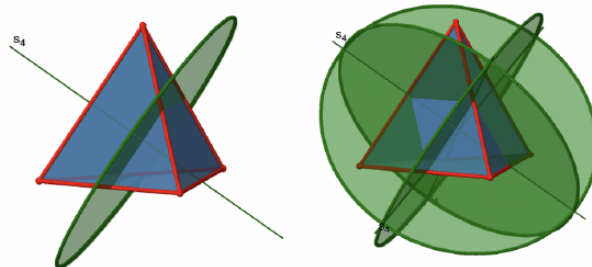


Figure 2.2.10 The S_4 axes in a tetrahedron (Attribution: symotter.org/gallery)

How many operations are associated with these S_4 axes? The order of the axes are even, and therefore we need four S_4 operations to produce the identity. The S_4^2 operation is the same as a C_2^1 operation because reflecting two times is equivalent to not reflecting at all, and rotating two times by 90° is the same as rotating about 180° . Therefore overall, only S_4^1 and S_4^3 operations are unique operations. S_4^2 and S_4^4 can be expressed by the simpler operations C_2^1 and E respectively. Because there are 3 S_4 axes, there are three S_4^1 and three S_4^3 operations. Overall there are six S_4 operations (Fig. 2.2.11).

Symmetry operations: $6S_4$

Symmetry operations in detail: $3 S_4^1, 3 S_4^3$

$$(S_4^4 = E, S_4^2 = C_2^1)$$

Figure 2.2.11 Symmetry operations associated with the S_4 axes in the point group T_d

There are also mirror planes (Fig. 2.2.11). The planes contain a single edge of the tetrahedron, thereby bisecting the tetrahedron. There are six edges in a tetrahedron, and therefore there are $6/1=6$ mirror planes.

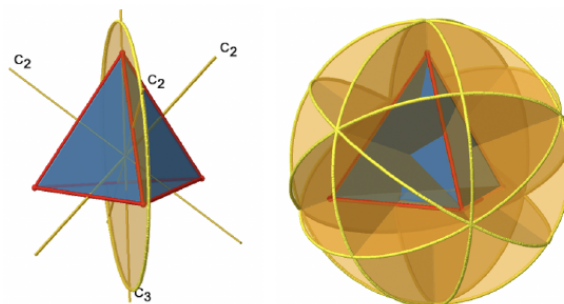


Figure 2.2.12 The mirror planes and C_2 axes in a tetrahedron belonging to the point group T_d (Attribution: symmotter.org/gallery)

These planes are dihedral planes because each plane contains a C_3 principal axis and is bisects the angle between two C_2 axes. Overall, there are three C_2 axes and three C_2 operations. There is one reflection operation per mirror plane because reflecting two times produces the identity. Therefore, there are six σ_d reflection operations (Fig. 2.2.12, right and Fig. 2.2.13).

Symmetry operations: $6\sigma_d$

Symmetry operations in detail: $6 \sigma_d$

Figure 2.2.13 Reflection operations associated with the σ_d mirror planes in T_d

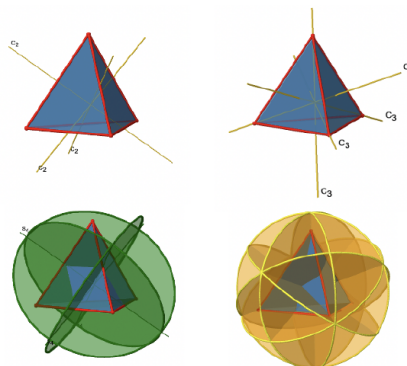


Figure 2.2.14 Summary of symmetry axes and planes in a tetrahedron with the point group T_d (Attribution: symmotter.org/gallery)

In sum (Fig. 2.2.14 and 2.2.15) we can denote the overall symmetry of the T_d point group the following way: $E, 8C_3, 3C_2, 6S_4, 6\sigma_d$. In detail the unique symmetry operations are $E, 4C_3^1, 4C_3^2, 3C_2^1, 3S_4^1, 3S_4^3, 6\sigma_d$.

Symmetry operations: $E, 8C_3, 3C_2, 6S_4, 6\sigma_d$

Symmetry operations in detail: $E, 4 C_3^1, 4 C_3^2, 3 C_2^1, 3 S_4^1, 3 S_4^3, 6 \sigma_d$
 $(C_3^3 = E, S_4^4 = E, S_4^2 = C_2^1)$

Figure 2.2.15 Summary of the symmetry operations for the point group T_d

The Rotational Subgroup T

The high symmetry point group T is the so-called rotational subgroup of the point group T_d . A rotational subgroup is a point group in which all symmetry operations but the identity and the proper rotations have been removed from a high-symmetry point group.

For the point group T this leaves the $4C_3$, the four C_3^2 and the three C_2 operations (Fig. 2.2.17). The S_4 rotation-reflections and the mirror planes have been removed. The point group T is rare.

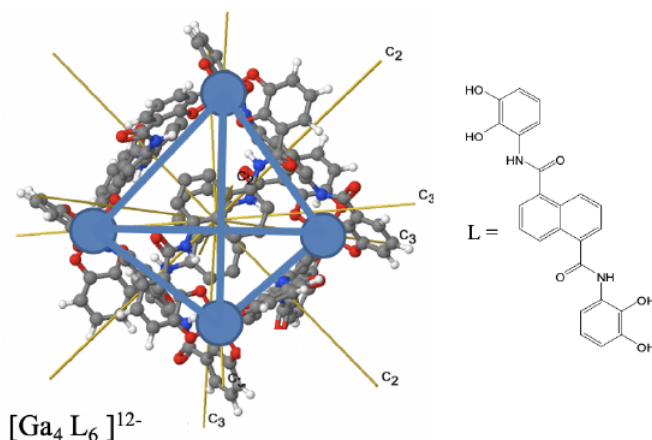


Figure 2.2.16 Symmetry elements in the Ga_4L_6 cage belonging to the point group T (Attribution: symmotter.org/gallery)

An example is the depicted Ga_4L_6 cage (Fig. 2.2.16). The Ga atoms occupy the vertices of a tetrahedron, but the point group is not T_d but T because of the shape of the ligands that connect the four Ga atoms.

Symmetry operations: E, $4C_3$, $4C_3^2$, $3C_2$

Figure 2.2.17 Symmetry operations for the Ga_4L_6 cage

The Octahedral Point Group O_h

Another high symmetry point group is the point group O_h . Both the octahedron as well as the cube belong to this point group despite their very different shape (Fig. 2.2.18). Because they belong to the same point group they must have the same symmetry elements and operations. There are many octahedrally shaped molecules, such as the SF_6 .

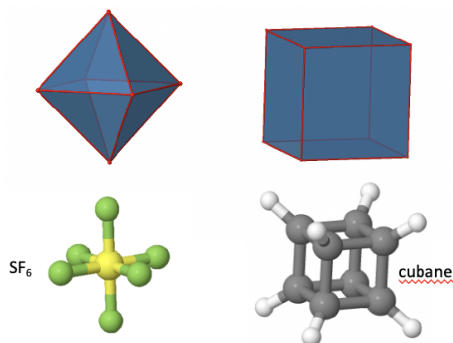


Figure 2.2.18 SF_6 and cubane with cubic and octahedral shape, respectively, belong to the point group O_h (Attribution: symmotter.org/gallery)

Molecules with cubic shapes are far less common, because a cubic shape often leads to significant strain in the molecule. An example is cubane C_8H_8 . Let us determine the symmetry elements and operations for the point group O_h using the example of the octahedron. If we used the cube, we would get exactly the same results.

There are three C_4 principal axes in the octahedron. They go through two opposite vertices of the octahedron (Fig. 2.2.19). There are three C_4 axes because an octahedron has six vertices which are all symmetry-equivalent because the octahedron is a platonic solid.

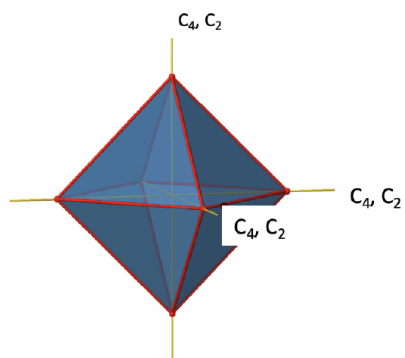


Figure 2.2.19 The C_4 and C_2 axes in the octahedral point group O_h (Attribution: symotter.org/gallery)

We can see that there are also C_2 axes where the C_4 axes run. This is because rotating two times around 90° is the same as rotating around 180° . What are the symmetry operations associated with these symmetry elements? Rotating four times around 90° using the C_4 axes produces the identity. So we have to consider the operations C_4^1 , C_4^2 , C_4^3 and C_4^4 . How many of these are unique? C_4^4 is the same as the identity, so it is not unique. In addition a C_4^2 is identical to a C_2^1 , and thus C_4^2 is also not unique, and can be expressed by the simpler operation C_2^1 . That leaves the C_4^1 and the C_4^3 as the only unique symmetry operations. Because we have three C_4 axes, there are $2 \times 3 = 6$ C_4 operations, in detail there are $3C_4^1$ and three C_4^3 operations. In addition, there are the three C_2^1 operations belonging to the three C_2 axes (Fig. 2.2.20).

Symmetry elements: E, $3C_4$, $3C_2$

Symmetry operations: E, $6C_4$, $3C_2$

In detail: E, $3C_4^1$, $3C_4^3$

($C_4^2 = C_2^1$, $C_4^4 = E$)

Figure 2.2.20 C_4 and C_2 symmetry operations in the octahedral point group O_h

In addition, there are four C_3 axes (Fig. 2.2.21). They are going through the center of two opposite triangular faces of the octahedron.

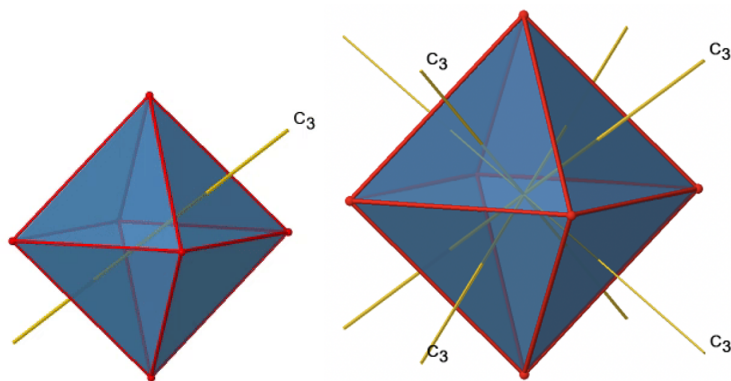


Figure 2.2.21 The C_3 axes in the octahedral point group O_h (Attribution: symotter.org/gallery)

You see above a single C_3 axis, and on the right hand side all four of these axes. How can we understand that there are four axes? An octahedron has overall eight triangular faces, and each C_3 axis goes through two opposite faces, so there are $8/2 = 4$ C_3 axes. Each C_3 axis has the C_3^1 and the C_3^2 as unique symmetry operations. The C_3^3 is the same as the identity. So overall we have $4 \times 2 = 8$ operations, four of them are C_3^1 , and four of them are C_3^2 (Fig. 2.2.22).

Symmetry elements: $4C_3$
 Symmetry operations: $8C_3$
 In detail: $4C_3^1, 4C_3^2$
 ($C_3^3 = E$)

Figure 2.2.22 Symmetry operations of C_3 in the octahedral point group O_h

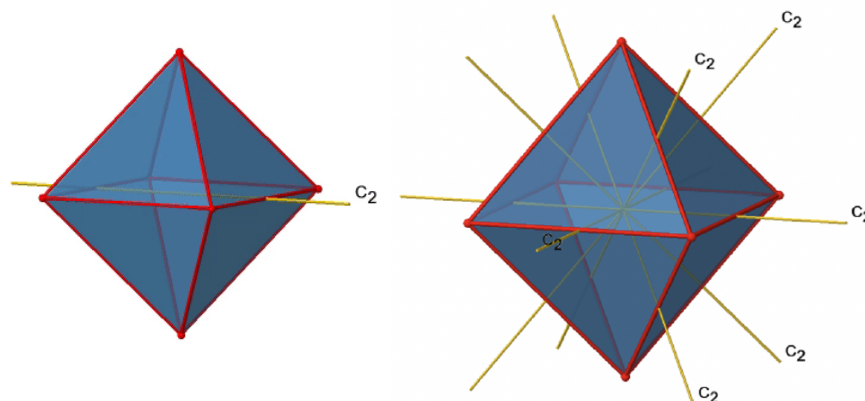


Figure 2.2.23 The C_2' axes in the octahedral point group O_h (Attribution: symotter.org/gallery)

In addition to the C_2 axes that superimpose the C_4 axes, there are C_2' axes which go through two opposite edges of the octahedron (Fig. 2.2.23). How many of them are there? An octahedron has twelve edges, and because each C_2' passes through two edges, there must be $12/2=6$ C_2' axes. These axes have primes because they are not conjugate to the C_2 axes that superimpose the C_4 axes. For each C_2' axis there is only the $C_2'^1$ as the unique symmetry operation, and therefore there are overall $6C_2'^1$ symmetry operations (Fig. 2.2.24).

Symmetry elements: $6C_2'$
 Symmetry operations: $6C_2'^1$

Figure 2.2.24 Symmetry operations of C_2' in the octahedral point group O_h

Let us look at the mirror planes next (Fig. 2.2.25). There are horizontal mirror planes that stand perpendicular to the C_4 principle axes. You can see a single one of them below on the left.

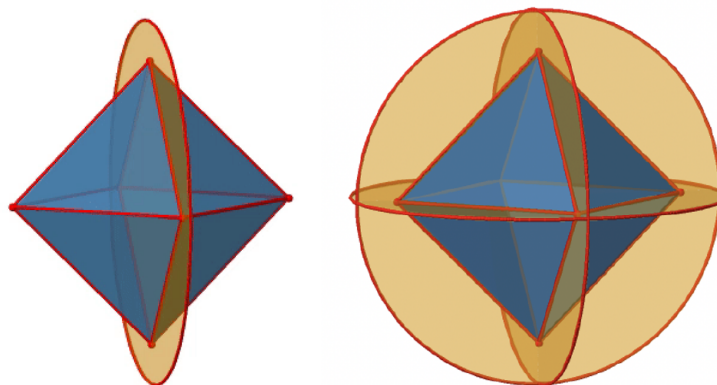


Figure 2.2.25 The horizontal mirror planes in the octahedral point group O_h (Attribution: symotter.org/gallery)

Note that this mirror plane also contains two axes, in addition to the one to which it stands perpendicular. Because it contains two principal C_4 axes, it has also properties of a vertical mirror plane. Nonetheless, we call it a horizontal mirror plane because it stands perpendicular to the third C_4 . The horizontal properties trump the vertical ones, so to say. You can see that a single mirror plane contains four edges of the octahedron. Because there are twelve edges, there are $12/4=3$ horizontal mirror planes. There is one

mirror plane per principal C_4 axis. There are three horizontal reflection operations because there is always only one reflection operation per mirror plane (Fig. 2.2.26).

Symmetry elements: $3\sigma_h$
Symmetry operations: $3\sigma_h$

Figure 2.2.26 Symmetry operations associated with horizontal mirror planes in the octahedral point group O_h

Next let us look for vertical mirror planes (Fig. 2.2.27). A vertical mirror plane is depicted below on the left.

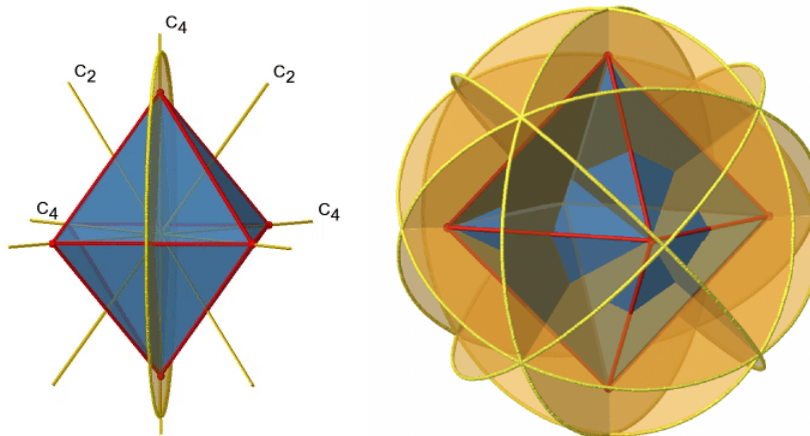


Figure 2.2.27 The vertical mirror planes in the octahedral point group O_h (Attribution: symmotter.org/gallery)

You can see that - contrast to the horizontal mirror planes - it does not contain any edges. Rather, it cuts through two opposite edges. You can see that this plane contains a C_4 axis, but it does not stand perpendicular to the other two C_4 axes. Therefore it has only the properties of a vertical mirror plane. You can see however, that the mirror plane bisects the angle between two C_2' axes which also depicted. This makes the vertical mirror planes dihedral mirror planes, σ_d . How many of them do we have? As previously mentioned, each mirror plane cuts through two opposite edges. There are twelve edges in an octahedron, and thus there are $12/2=6$ dihedral mirror planes. You can see all of them on the right side of Fig. 2.2.27. Each mirror plane is associated with one reflection operation, therefore there are six dihedral reflection operations (Fig. 2.2.28).

Symmetry elements: $6\sigma_d$
Symmetry operations: $6\sigma_d$

Figure 2.2.28 Symmetry operations associated with vertical mirror planes in the octahedral point group O_h

Next we can ask if the point group O_h has an inversion center? Yes, there is one in the center of the octahedron (Fig. 2.2.29)!

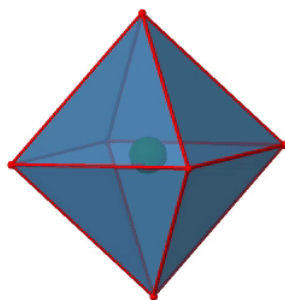


Figure 2.2.29 The inversion center of the octahedral point group O_h (Attribution: symmotter.org/gallery)

Each point in the octahedron can be moved through the inversion center to the other side, and the produced octahedron will superimpose the original one. There is always one inversion operation associated with an inversion center (Fig. 2.2.30).

Symmetry elements: i
Symmetry operations: i

Figure 2.2.30 Symmetry operations associated with the inversion center in the octahedral point group

Next, let us look for rotation-reflections. You can see an S_6 rotation-reflection operation below (Fig. 2.2.31, left).

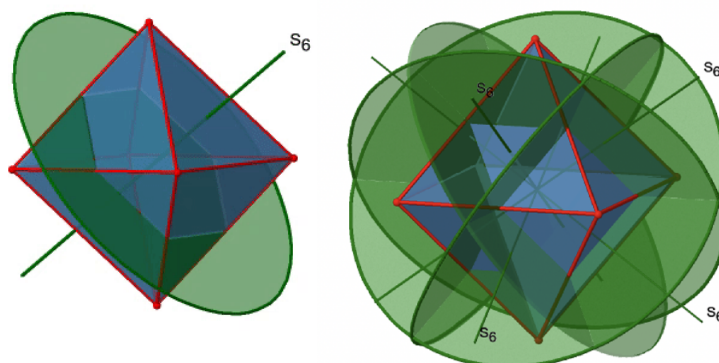


Figure 2.2.31 The S_6 rotation-reflection element of the octahedral point group O_h (Attribution: symmotter.org/gallery)

The improper S_6 axis passes through the centers of two opposite triangular faces. One can see that rotation about 60° alone does not make the octahedron superimpose. The reflection at a plane perpendicular to the improper axis is required to achieve superposition. Overall, the rotation-reflection swaps up the position of the two opposite triangular faces. How many S_6 improper axes are there? Since each S_6 passes through two faces, and an octahedron has 8 faces there must be $8/2=4$ S_6 axes. You can see all of them above (Fig. 2.2.31, right). Note that they are in the same position as the $4C_3$ axes we previously discussed. How many unique operations are associated with them? For an S_6 axis we need to consider operations from S_6^1 to S_6^6 . S_6^6 is the same as the identity so it is not unique. The S_6^2 is the same as a C_3^1 because rotating two times round 60° is the same as rotating around 120° , and reflecting twice is the same as not reflecting at all. Similarly, an S_6^4 is the same as a C_3^2 . Rotating four times by 60° is the same as rotating two times by 120° and reflecting four times is the same as not reflecting at all. Further, an S_6^3 is the same as an inversion. After three 60° rotations we have rotated by 180° . If we reflect after that, then this is the same as an S_2^1 operation which is the same as an inversion. Therefore, only the S_6^1 and the S_6^5 operations are unique, all other operations can be expressed by simpler operations (Fig. 2.2.32).

Symmetry elements: 4 S_6
Symmetry operations: 8 S_6
In detail $4S_6^1, 4S_6^5$
($S_6^2=C_3^1, S_6^3=i, S_6^4=C_3^2, S_6^6=E$)

Figure 2.2.32 Symmetry operations of the rotation-reflection S_6 of the octahedral point group O_h

The octahedron also has S_4 improper axes, and you can see one of them below (Fig. 2.2.33, right).

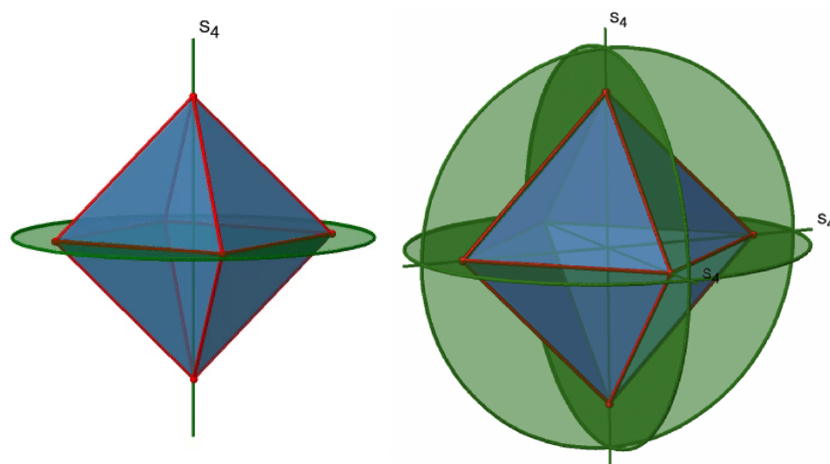


Figure 2.2.33 The S_4 improper axis of the octahedral point group O_h (Attribution: symmotter.org/gallery)

It goes through two opposite corners of the octahedron. The S_4 improper axis seemingly does the same as the C_4 axis that goes through the same two opposite vertices, but actually does not. While rotating around 90° already makes the octahedron superimpose with its original form, executing the reflection operation after the rotation swaps up the position of the two vertices, and generally all points of the octahedron above and below the plane, respectively. Overall the S_4 moves the points within the object differently compared to the C_4 which makes it an additional, unique symmetry element. There are overall three S_4 improper axes because the octahedron has six vertices and one S_4 passes through two vertices (Fig. 2.2.34).

Symmetry elements: 3 S_4
 Symmetry operations: 6 S_4
 In detail: $3S_4^1, 3S_4^3$
 ($S_4^2=C_2$, $S_4^4=E$)

Figure 2.2.34 Symmetry operations associated with the S_4 improper axis of the octahedral point group O_h

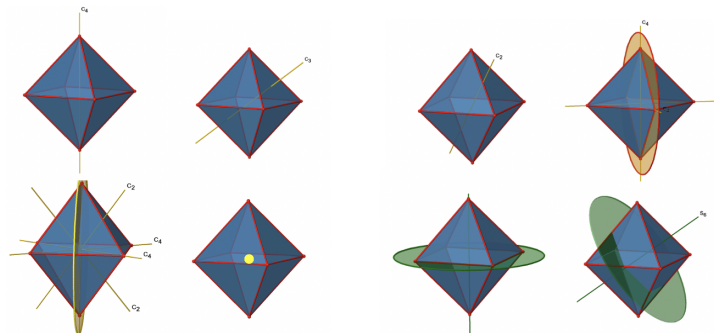


Figure 2.2.35 An overview over all symmetry elements in the octahedral point group O_h (Attribution: symmotter.org/gallery)

Here is an overview over all the symmetry elements and operations (Fig. 2.2.35 and 2.2.36). Overall, there are 48 different unique operations that one can perform!

Symmetry elements: $E, i, 4S_6, 4C_3, 3S_4, 3C_4, 6C_2, 3C_2, 3\sigma_h, 6\sigma_d$
 Symmetry operations: $E, i, 8S_6, 8C_3, 6S_4, 6C_4, 6C_2, 3C_2, 3\sigma_h, 6\sigma_d$
 In detail: $E, i, 4S_6^1, 4S_6^5, 4C_3^1, 4C_3^2, 3S_4^1, 3S_4^3, 3C_4^1, 3C_4^3, 6C_2, 3C_2, 3\sigma_h, 6\sigma_d$
 ($S_6^2=C_3^1, S_6^3=i, S_6^4=C_3^2, S_6^5=E, C_3^3=E, S_4^2=C_2^1, S_4^4=E, C_4^2=C_2^1, C_4^4=E$)

Figure 2.2.36 Summary over all symmetry operations and elements of the octahedral point group O_h

Like the point group T_d , also the point group O_h has a rotational subgroup, named O . It has the identity and the same proper rotations as the point group O_h , but no other symmetry operations (Fig. 2.2.38). An example is the polyoxometalate cluster core shown below (Fig. 2.2.37). Polyoxometalates are cluster anions of the group 5 and 6 elements.

Figure 2.2.37 Proper rotations in a $V_6P_8O_{24}$ polyoxometalate cluster core. The C_3 rotation is animated. (Attribution: symmotter.org/gallery)

The point group O is generally rare.

Symmetry elements: $E, i, 4S_6, 4C_3, 3S_4, 3C_4, 6C_2, 3C_2, 3\sigma_d, 6\sigma_d$
 Symmetry operations: $E, i, 8S_6, 8C_3, 6S_4, 6C_4, 6C_2, 3C_2, 3\sigma_d, 6\sigma_d$
 In detail: $E, i, 4S_6, 4S_6^5, 4C_3, 4C_3^2, 3S_4, 3S_4^3, 3C_4, 3C_4^3, 6C_2, 3C_2, 3\sigma_d, 6\sigma_d$

Figure 2.2.38 Symmetry elements and operations in the point group O (relative to O_h)

Another high symmetry point group is the point group T_h . It can also be derived from the point group O_h . In this case the S_4 , the C_4 , the C_2' , and the σ_d operations are removed from the octahedral symmetry. An example is the hexapyridyl iron (2+) cation (Fig. 2.2.39).

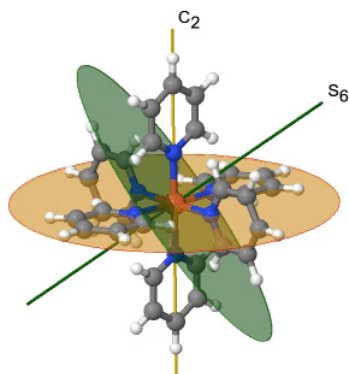


Figure 2.2.39 The hexapyridyl iron (2+) cation and its S_6 and C_2 symmetry elements. An S_6 operation is animated.

You can see that the N-atoms of the pyridyl-ligands surround the Fe atoms octahedrally, but the symmetry is reduced from O_h to T_h because of the planar shape of the pyridyl-ligands. In particular the C_4 symmetry is reduced to C_2 . This reduction in symmetry leads to elimination of the S_4 , the C_2' , and the σ_d symmetry elements (Fig. 2.2.40).

T_h group (symmetry elements: $E, i, 4S_6, 4C_3, 3C_2, 3\sigma_h$) can also be considered as a result of reducing O_h group symmetry ($E, i, 4S_6, 4C_3, 3S_4, 3C_4, 6C_2, 3C_2, 3\sigma_h, 6\sigma_d$) by eliminating C_4 , S_4 and some C_2 axes and σ_d planes

Symmetry operations: $i, 8S_6, 8C_3, 3C_2, 3\sigma_h$

Figure 2.2.40 Symmetry operations in $Fe(py)_6^{2+}$ in the point group T_h

The I_h Point Group

The two remaining platonic solids, the icosahedron and the dodecahedron, belong both to the icosahedral point group I_h . This is despite they are made of different polygons (Fig. 2.2.41).

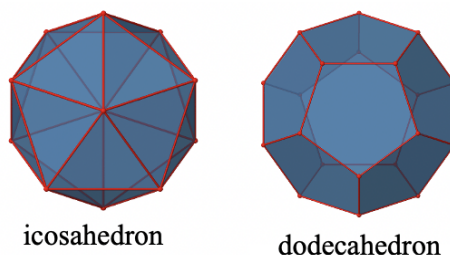


Figure 2.2.41 The icosahedron and dodecahedron of the I_h point group (Attribution: symmotter.org/gallery)

Because they belong to the same point group, they have exactly the same symmetry operations. An example for a molecule with icosahedral shape is the molecular anion $B_{12}H_{12}^{2-}$. An example for a molecule with dodecahedral shape is the dodecahedrane $C_{20}H_{20}$.

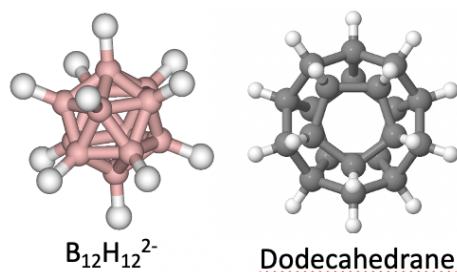


Figure 2.2.42 $B_{12}H_{12}^{2-}$ and the dodecahedrane (Attribution: wikiwand.com <https://upload.wikimedia.org/Wikipedia/commons/thumb/9/97/Dodecaborane-3D-balls.png/640px-Dodecaborane-3D-balls.png>)

Let us determine the symmetry elements and symmetry operations for the example of the icosahedron. We could also use the dodecahedron, and the results would be the same. The principal axes of the icosahedron are the C_5 axes. You can see one of them, going through the center of the pentagon comprised of five triangular faces below (Fig. 2.2.43).

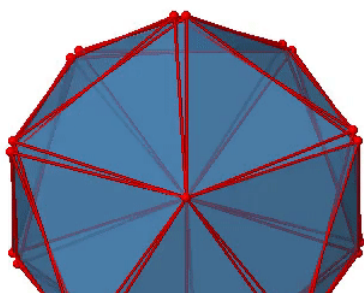


Figure 2.2.43 One of the C_5 axes of the icosahedron stands perpendicular to the paper plane going through the center of a pentagon of the icosahedron (Attribution: symotter.org/gallery)

You can understand that there is a C_5 when considering that there are five triangular faces making a pentagon. The C_5 axis sits in the center of the pentagon. We can see that when we rotate around this C_5 axis, then the produced icosahedron superimposes the original one. The C_5 axis goes through two opposite vertices of the icosahedron. Because an icosahedron has 12 vertices, there must be six C_5 axes overall. You can see all of them below (Fig. 2.2.44).

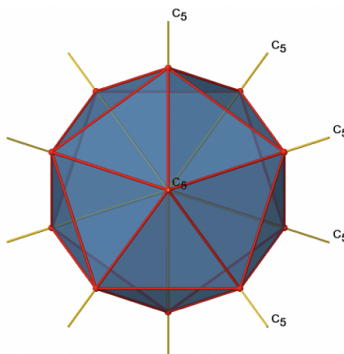


Figure 2.2.44 The C_5 axes of the icosahedron (Attribution: symotter.org/gallery)

There are four unique symmetry operations associated with a single C_5 axis, namely the C_5^1 , the C_5^2 , the C_5^3 , and the C_5^4 . The C_5^5 is the same as the identity. Because there are six C_5 axes, there are overall $6 \times 4 = 24$ C_5 symmetry operations (Fig. 2.2.45).

Symmetry elements: 6 C_5
 Symmetry operations: 24 C_5
 In detail: 6 C_5^1 , 6 C_5^2 , 6 C_5^3 , 6 C_5^4 ($C_5^5=E$)

Figure 2.2.45 Symmetry operations associated with the C_5 axis of icosahedral point group

In addition, there are C_3 axes. One of them is shown below, and you can see that it passes through the centers of two opposite triangular faces (Fig. 2.2.46).

Figure 2.2.46 One of the C_3 axes of the icosahedral point group (Attribution: symmotter.org/gallery)

As one rotates by 120° the atoms on the triangular faces change their position, and the resulting icosahedron superimposes the original one. As the name icosahedron says, there are twenty faces overall.

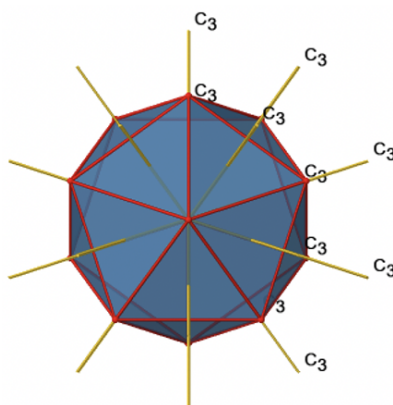


Figure 2.2.47 The C_3 axes of the icosahedral point group (Attribution: symmotter.org/gallery)

Because one C_3 passes through two opposite axes, there are $20/2=10$ C_3 axes overall (Fig. 2.2.47). Each C_3 axis is associated with two symmetry operations, namely C_3^1 , and C_3^2 . Thus, there are overall $10 \times 2 = 20$ C_3 symmetry operations.

Symmetry elements: 10 C_3
 Symmetry operations: 20 C_3
 In detail: 10 C_3^1 , 10 C_3^2 ($C_3^3=E$)

Figure 2.2.48 Symmetry operations of the C_3 axes of the icosahedral point group

There are also C_2 axes (Fig. 2.2.49). They pass through the centers of two opposite edges of the icosahedron. Rotating around the C_2 axis shown makes the icosahedron superimpose.

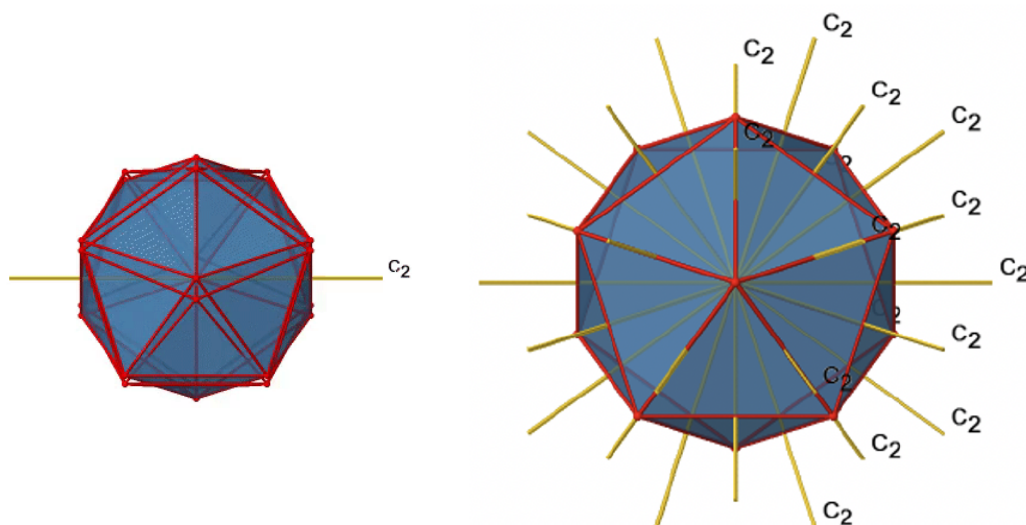


Figure 2.2.49 The C_2 axes of the icosahedral point group (Attribution: symotter.org/gallery)

An icosahedron has overall 30 edges. Because one C_2 axis passes through the centers of two opposite edges, we can understand that there are $30/2=15$ C_2 axes. There is one unique C_2 operation per axis, and therefore there are 15 C_2 operations (Fig. 2.2.50).

Symmetry elements: 15 C_2
 Symmetry operations: 15 C_2'
 In detail: 15 $C_2'^1$

Figure 2.2.50 Symmetry operations associated with the C_2 axes of the icosahedral point group

We have now found all proper rotations. Let us look for mirror planes, next. You can see a mirror plane below (Fig. 2.2.51).

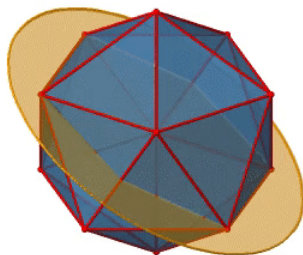


Figure 2.2.51 A mirror plane in the icosahedral point group (Attribution: symotter.org/gallery)

It contains two opposite edges. It also bisects two other edges. An icosahedron has overall 30 edges, therefore there are $30/2=15$ mirror planes. You can see all of them below (Fig. 2.2.52 and Fig. 2.2.53)).

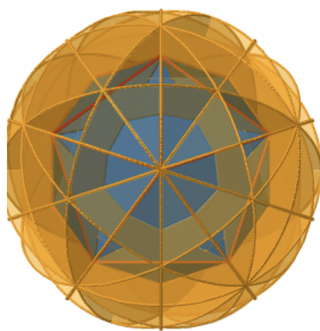


Figure 2.2.52 All the mirror planes in the icosahedral point group (Attribution: symmotter.org/gallery)

Symmetry elements: 15σ
Symmetry operations: 15σ

Figure 2.2.53 Symmetry operations of the mirror planes in the icosahedral point group

The icosahedron also has an inversion center in the center of the icosahedron (Fig. 2.2.54 and Fig. 2.2.55)).

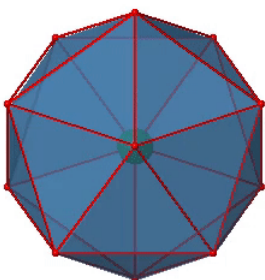


Figure 2.2.54 The inversion center in the icosahedron (Attribution: symmotter.org/gallery)

As we carry out the associated, one symmetry operation, all points in the icosahedron move through the inversion center to the other side.

Symmetry elements: i
Symmetry operations: i

Figure 2.2.55 Symmetry operations of the inversion planes in the icosahedral point group

Let us now look for improper rotations. The improper rotational axes with the highest order are S_{10} axes. They are located in the same position as the C_5 axes, and go through two opposite corners (Fig. 2.2.56).

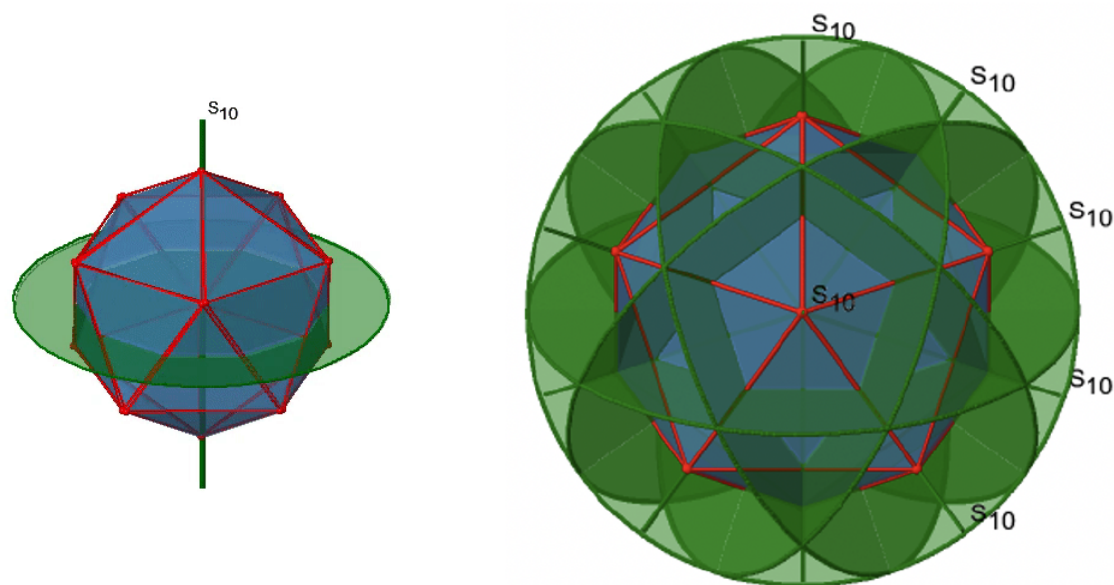


Figure 2.2.56 The S_{10} improper rotational axes in the icosahedral point group (Attribution: symmotter.org/gallery)

The S_{10} exists because in an icosahedron there are pairs of co-planar pentagons that are oriented staggered relative to each other. The rotation around 36° brings one pentagon in eclipsed position relative to the other, but superposition is only achieved after the reflection at the mirror plane perpendicular to the rotational axis. Because one S_{10} passes through two opposite vertices, and there are 12 vertices there are 6 S_{10} improper axes. For each axis there are four unique symmetry operations, the S_{10}^1 , the S_{10}^3 , the S_{10}^7 , and the S_{10}^9 . Therefore, there are overall $4 \times 6 = 24$ operations possible (Fig. 2.2.57).

Symmetry elements: 6 S_{10}

Symmetry operations: 24 S_{10}

In detail: 6 S_{10}^1 , 6 S_{10}^3 , 6 S_{10}^7 , 6 S_{10}^9 ($S_{10}^2 = C_5^1$, $S_{10}^4 = C_5^2$, $S_{10}^5 = i$, $S_{10}^6 = C_5^3$, $S_{10}^8 = C_5^4$)

Figure 2.2.57 Symmetry operations associated with the S_{10} improper rotational axes of the icosahedral point group

Are the lower order improper rotational axes? Yes, there are S_6 axes that pass through the centers of two opposite triangular faces (Fig. 2.2.58). This symmetry element exists because the two triangular faces are in staggered orientation to each other. Rotation alone brings one face in eclipsed orientation relative to the other, but reflection at a mirror plane perpendicular to the axis is required to achieve superposition. The S_6 axes are in the same location as the C_3 axes.

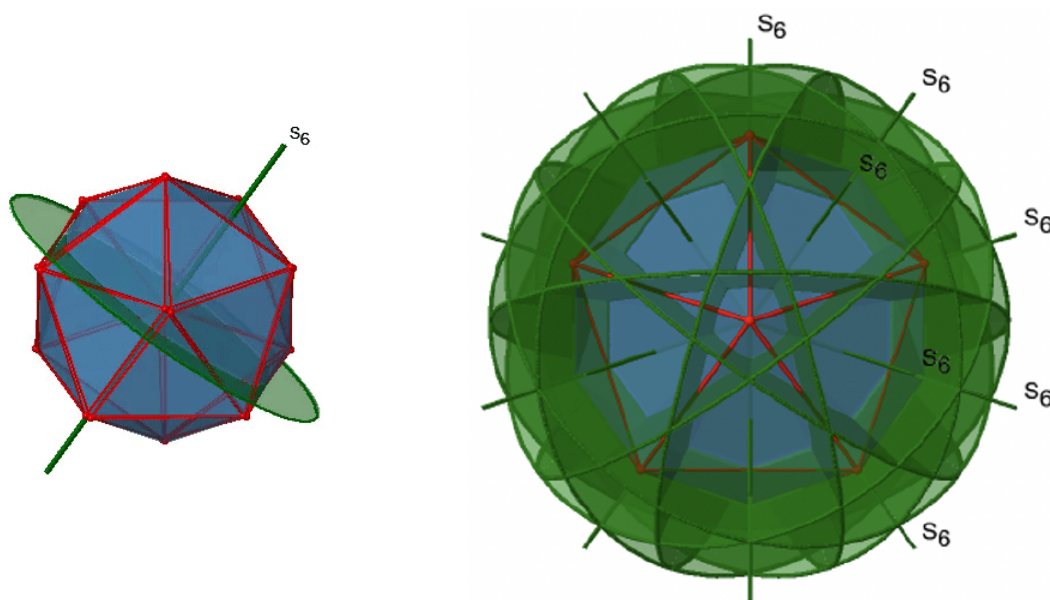


Figure 2.2.58 The S_6 rotation-reflections in the icosahedral point group (Attribution: symmotter.org/gallery)

There are 10 S_6 axes because there are twenty faces and one axis passes through two opposite faces. Only the S_6^1 and the S_6^5 operations are unique S_6 operations, all others can be expressed by simpler operations. Therefore there are overall $10 S_6^1 + 10 S_6^5 = 20 S_6$ operations (Fig. 2.2.59).

Symmetry elements: 10 S_6
 Symmetry operations: 20 S_6
 In detail: $10 S_6^1$, $10 S_6^5$ ($S_6^2 = C_3^1$, $S_6^3 = i$, $S_6^4 = C_3^2$)

Figure 2.2.59 Symmetry operations associated with the S_6 rotation-reflections in the icosahedral point group

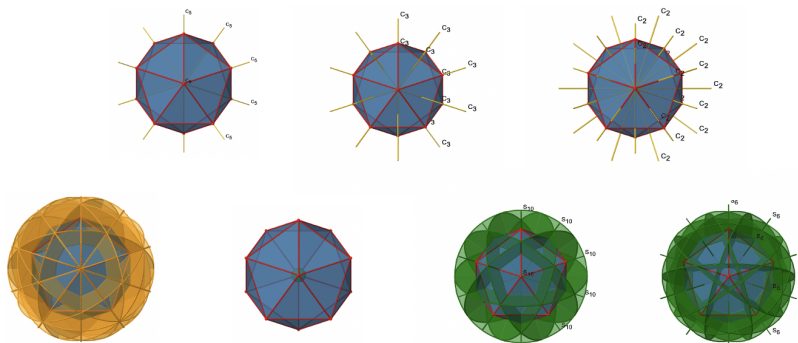


Figure 2.2.60 The symmetry elements of the icosahedral point group (Attribution: symotter.org/gallery)

We have now found all symmetry operations for the I_h symmetry. There are overall 120 operations making the point group I_h the point group with the highest symmetry (Fig. 2.2.60 and Fig. 2.2.61).

Symmetry elements: E, 6 C_5 , 10 C_3 , 15 C_2 , 15 σ , i, 6 S_{10} , 10 S_6
 Symmetry operations: E, 24 C_5 , 20 C_3 , 30 C_2 , 15 σ , i, 24 S_{10} , 20 S_6
 In detail: E, 6 C_5^1 , 6 C_5^2 , 6 C_5^3 , 6 C_5^4 , 10 C_3^1 , 10 C_3^2 , 30 C_2^1 , 15 σ , i, 6 S_{10}^1 , 6 S_{10}^3 , 6 S_{10}^7 , 6 S_{10}^9 ,
 $10 S_6^1$, $10 S_6^5$

Figure 2.2.61 Summary of the symmetry operations of the icosahedral point group

Also the point group I_h has a rotational subgroup.

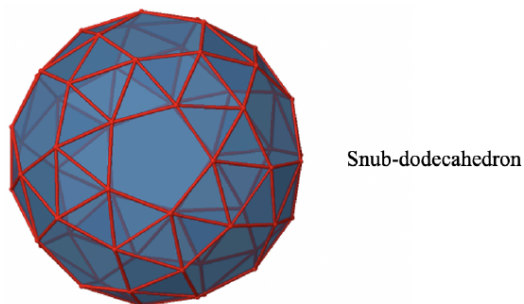


Figure 2.2.62 The snub-dodecahedron belonging to the point group I (Attribution: symotter.org/gallery)

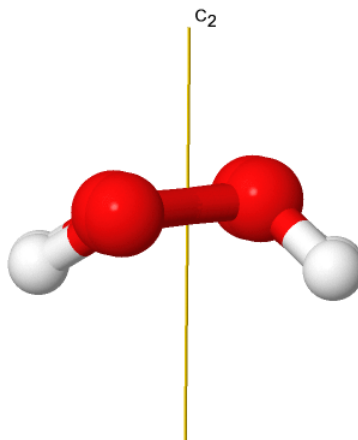
It is called I. An example of an object with this symmetry is the snub-dodecahedron (Fig. 2.2.62). It has the identity, and all the proper rotation operations of the point group O_h , but the inversion, the rotation-reflections, and the mirror planes are eliminated (Fig. 2.2.63).

Symmetry elements: E, 6 C_5 , 10 C_3 , 15 C_2 , ~~15 σ , i, 6 S_{10} , 10 S_6~~
 Symmetry operations: E, 24 C_5 , 20 C_3 , 30 C_2 , ~~15 σ , i, 24 S_{10} , 20 S_6~~
 In detail: E, 6 C_5^1 , 6 C_5^2 , 6 C_5^3 , 6 C_5^4 , 10 C_3^1 , 10 C_3^2 , 30 C_2^1 , ~~15 σ , i, 6 S_{10}^1 , 6 S_{10}^3 , 6 S_{10}^7 , 6 S_{10}^9 ,
 $10 S_6^1$, $10 S_6^5$~~

Figure 2.2.63 Symmetry operations of the icosahedral rotational sub group I (relative to I_h)

Cyclic Point Groups

After having discussed high and low symmetry point groups, let us next look at cyclic point groups. They have the property that they have only a single proper n-fold rotational axis, but no other proper axes. In the most simple case they do not have any additional symmetry element such as mirror planes or rotation-reflections. These point groups are denoted C_n whereby n is the order of the proper axis. An example is the hydrogen peroxide molecule H_2O_2 (Fig. 2.2.64).



Point Group = C_2 , Order = 2

Figure 2.2.64 The C_2 rotational axis of hydrogen peroxide

It has a so-called roof-structure due to its non-planarity. One hydrogen atom points toward us, and the other points away from us. This structure is due to the two electron-lone pairs at each sp^3 -hybridized oxygen atom. These electron-lone pairs consume somewhat more space than the H atoms, and there is electrostatic repulsion between the electron lone pairs. Therefore, the electron lone pairs at the different oxygen atoms try to achieve the greatest distance from each other. This forces the H-atoms out of the plane, leading to the roof-structure of the hydrogen peroxide. Because the H_2O_2 molecule is not planar, it only has a single C_2 axis, but no other symmetry element besides the identity. The C_2 axis passes through the center of the O-O bond. Execution of the C_2 operation swaps up both the O and the H atoms.

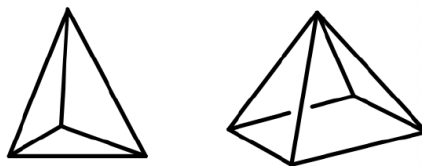
$C_2 - H_2O_2$ (elements: E, C_2)

Figure 2.2.65 Symmetry elements in the hydrogen peroxide molecule

Definition: Cyclic Groups C_n

Cyclic groups have one rotational axis

Another class of groups are the pyramidal groups, denoted C_{nv} . They have n vertical mirror planes containing the principal axis C_n in addition to the principal axis C_n . Generally molecules belonging to pyramidal groups are derived from an n -gonal pyramid. An n -gonal pyramid has an n -gonal polygon as the basis which is capped (Fig. 2.2.66).



N-gonal pyramid

Figure 2.2.66 n -gonal pyramids ($n=3,4$)

For example a trigonal pyramid has a triangular basis which is capped, a tetragonal pyramid has a square which is capped, and so on. The proper axis associated with a specific pyramid has the order n and goes through the tip of the pyramid and the center of the polygon. An example of a molecule with a trigonal pyramidal shape is the NH_3 (Fig. 2.2.67).

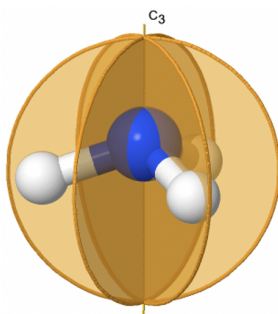


Figure 2.2.67 C_3 axis and vertical mirror planes in NH_3

The three H atoms form the triangular basis of the pyramid, which is capped by the N atom. The NH_3 molecule belongs to the point group C_{3v} . The C_3 axis goes through the N atom which is the tip of the pyramid, and the center of the triangle defined by the H atoms. There are three vertical mirror planes that contain the C_3 axis. Each of them goes through an N-H bond (Fig. 2.2.68).

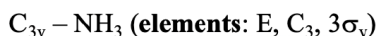


Figure 2.2.68 Symmetry elements in NH_3

Definition: Pyramidal Groups C_{nv}

Pyramidal groups have n vertical plane(s) in addition to the principal axis C_n .

The Linear Group $C_{\infty v}$

A special n -gonal polygon is the cone. A cone can be conceived as an n -gonal pyramid with an infinite number n of corners at the base (Fig. 2.2.69).

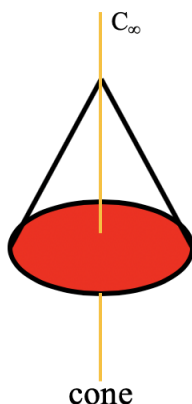


Figure 2.2.69 Cone having a proper rotational axis with infinite order.

In this case the order of the rotational axis that passes through the tip of the cone and the center of the circular basis is infinite. This also means that there is an infinite number of vertical mirror planes that contain the C_{∞} axis (Fig. 2.2.71). The point group describing the symmetry of a cone is called the linear point group $C_{\infty v}$. Polar, linear molecules such as CO, HF, N_2O , and HCN belong to this point group. You can see the HCN molecule with its C_{∞} axis and its infinite number of vertical mirror planes below (Fig. 2.2.70).

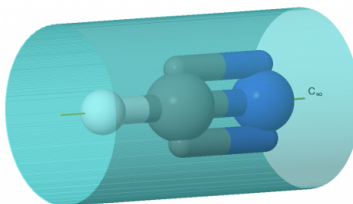


Figure 2.2.70 C_{∞} axis of the HCN molecule

The infinite number of mirror planes, shown in blue are forming a cylinder that surround the molecule.

(elements: $E, C_{\infty}, \infty \sigma_v$)

Figure 2.2.71 Symmetry elements of polar linear molecules

Definition: Linear group $C_{\infty v}$

The linear group $C_{\infty v}$ has an infinite number of vertical mirror planes containing a C_{∞} axis

Reflection Groups

If we add a horizontal mirror plane instead of n vertical mirror planes to a proper rotational axis C_n we arrive at a reflection point group type C_{nh} . The presence of the horizontal mirror planes also generates an improper axis of the order n . This is because when one can rotate and reflect perpendicular to the rotational axes independently, then it must also be possible to do it in combination. An example of a molecule belong to a reflection group is the trans-difluorodiazene N_2F_2 (Fig. 2.2.72).

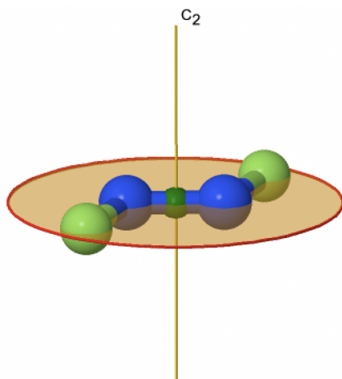


Figure 2.2.72 C_2 axis and horizontal mirror plane in trans- N_2F_2

It is a planar molecule with a C_2 axis going through the middle of the N-N double bond, and standing perpendicular to the plane of the molecule. The horizontal mirror plane stands perpendicular to the C_2 axis, and is within the plane of the molecule. There is an additional inversion center because an S_2 must exist which is the same as an inversion center. The inversion center is in the middle of the N-N bonds. Overall, the molecule has the symmetry C_{2h} .

C_{2h} – trans-F-N=N-F (elements: E, C_2, i, σ_h)

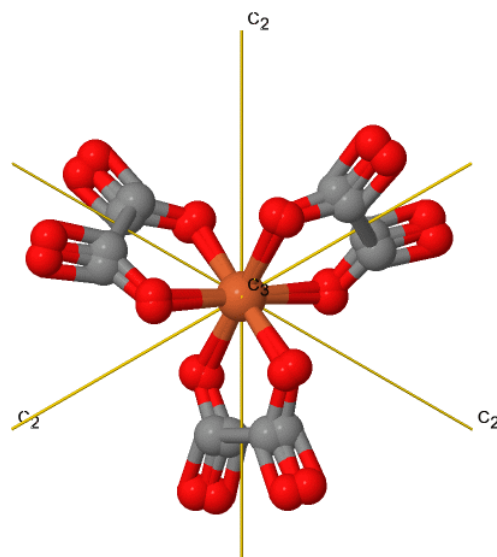
Figure 2.2.73 Symmetry elements in N_2F_2

Definition: Reflection Group C_{nh}

A reflection group has a horizontal plane perpendicular to the principal axis C_n

Dihedral Groups

Dihedral groups are point groups that have n additional C_2 axes that stand perpendicular to the principal axis of the order n . If there are no other symmetry elements, then the point group is of the type D_n .



Point Group = D_3 , Order = 6

Figure 2.2.74 The tris-oxalato ferrate (3-) ion and its symmetry elements

For example in the point group D_3 there is a C_3 principal axis, and three additional C_2 axes, but no other symmetry element (Fig. 2.2.75). The tris-oxalato ferrate (3-) ion belongs to this point group (Fig. 2.2.75). You can see that the C_3 axis stands perpendicular to the paper plane, and there are three C_2 axes in the paper plane.

D_3 – (elements: E, C_3 , $3C_2$)

Figure 2.2.75 Symmetry elements in the dihedral group D_3

Definition: Dihedral Groups D_n

In a point group of the type D_n there is a principal axis of order n , n C_2 axes, but no other symmetry elements.

If a horizontal mirror plane is added to the C_n axis and the n C_2 axes we arrive at the prismatic point groups D_{nh} (Fig. 2.2.76). The addition of the horizontal mirror plane generates further symmetry elements namely an S_n and n vertical mirror planes.

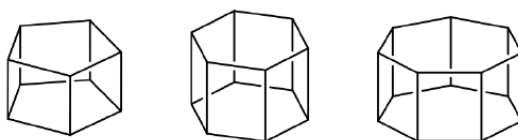


Figure 2.2.76 N-gonal prisms belong to the point group type D_{nh}

Generally, molecules belonging to this point group derive from n-gonal prisms. The order of the principal axis is the same as the number of corners of the polygons the prism are made of.

Definition: Prismatic Groups D_{nh}

In prismatic point groups there is a horizontal mirror plane perpendicular to the principal axis C_n . There are also n C_2 axes.

An example for a molecule belonging to a prismatic point group is PF_5 (Fig. 2.2.77).

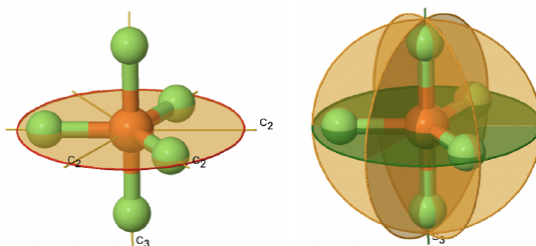


Figure 2.2.77 The PF_5 molecule belonging to the point group D_{3h} and its symmetry elements.

It has a trigonal bipyramidal shape. The C_3 axis goes through the axial F atoms of the molecule, and the three C_2 axes go through the three equatorial F atom. The horizontal mirror plane stands perpendicular to the principal C_3 axis and is located within the equatorial plane of the molecule. In addition, there are the vertical mirror planes that contain the C_3 axis, and go through the three equatorial P-F bonds. There is also an S_3 axis which superimposes the C_3 axis. In sum:

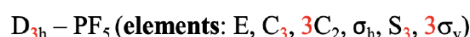


Figure 2.2.78 Symmetry elements of PF_5

A special case of a D_{nh} group is the linear group $D_{\infty h}$. An object that has this symmetry is a cylinder. A cylinder can be conceived as a prism with an infinite number of vertices. Thus, the principal axis that passes through a cylinder has infinite order. Because of the infinite order of the principal axis, there is an infinite number of C_2 axes that stand perpendicular to the principal axis. You can see one such C_2 going through the cylinder (Fig. 2.2.79).

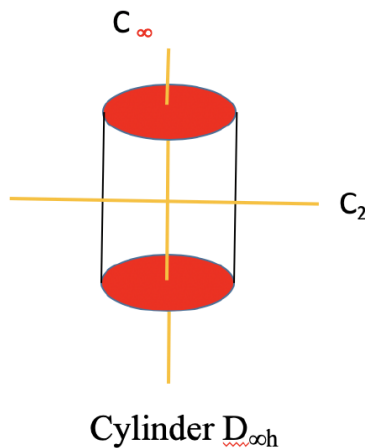


Figure 2.2.79 Cylinder as an example of linear group $D_{\infty h}$

There is now also an improper axis of infinite order, as well as an infinite number of vertical mirror planes. Non-polar linear molecules like H_2 , CO_2 , and acetylene C_2H_2 belong to the point group $D_{\infty h}$. You can see the C_{∞} axis passing through a CO_2 molecule below (Fig. 2.2.80).

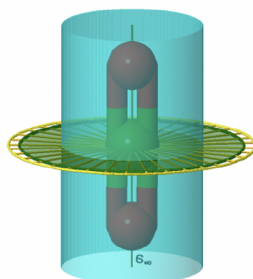


Figure 2.2.80 C_∞ in a CO₂ molecule

You can see the infinite number of vertical mirror planes as a blue cylinder. The infinite number of C₂ axes is shown a yellow lines going around the molecule. In sum:

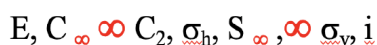


Figure 2.2.81 Symmetry elements of linear group D_{∞h}

Definition: Linear Group D_{∞h}

In the point group D_{∞h} there is an infinite number of n C₂ axes in addition to the principal axis of infinite order, an infinite number of vertical mirror planes, and one horizontal mirror plane.

If we add n vertical mirror planes to the principal axis and the n C₂ axes, we arrive at the point group D_{nd}. The vertical mirror planes are dihedral mirror planes because they bisect the angle between the C₂ axes. An example is the ethane molecule in staggered conformation which has the symmetry D_{3d} (Fig. 2.2.82).

Figure 2.2.1: Paste Caption Here

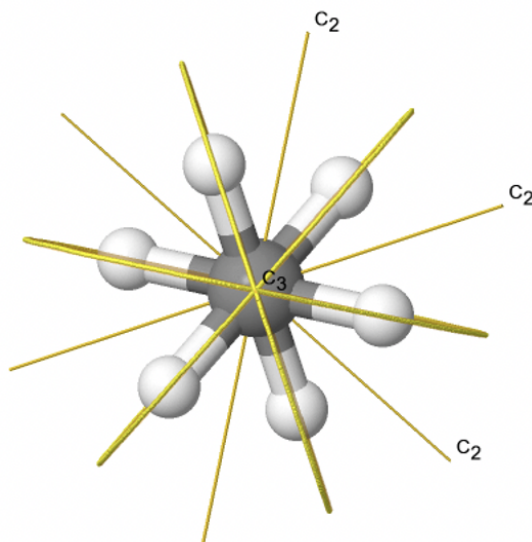


Figure 2.2.82 The ethane molecule in the staggered conformation belongs to the point group type D_{nd}

The C₃ axis goes along the C-C bond, and the 3C₂ axes pass through the middle of the carbon-carbon bond, and bisect the angle between two hydrogens and one carbon atom. The three dihedral mirror planes pass through the C-H bonds. In addition, the ethane molecule has an S₆ axis, and an inversion center. In sum:

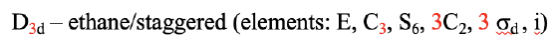


Figure 2.2.83 Symmetry elements in ethane

Definition: D_{nd}

In this point group type there are n dihedral mirror planes that contain the C_n and bisect the angle between adjacent C_2 axes

Improper Rotational Point Groups

The last class of point groups to be discussed are the improper rotation point groups. They only have one proper rotational axis, and an improper rotational axis that has twice the order of the proper rotational axis (Fig. 2.2.85). There may be an inversion center present depending on the order of the proper and improper axes. An example is the tetramethylcycloocta-tetraene molecule (Fig. 2.2.84).

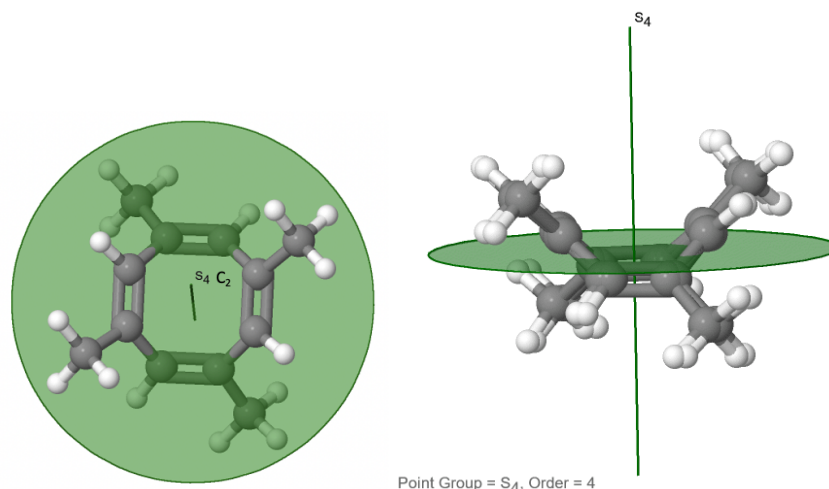


Figure 2.2.84 The S_4 and an C_2 axes of tetramethyl cycloocta-tetraene

It has an S_4 and an C_2 axis as the only symmetry elements besides the identity. Rotating by 90° alone does not superimpose the molecule because two C-C double bonds lie above the plane and two below the plane. In addition, two opposite methyl groups lie above and below the plane respectively. Therefore it needs the additional reflection to achieve superposition. There is also a C_2 axis which is in the same locations as the S_4 axis.

Symmetry elements: E, S_{2n} , C_n

May have inversion center i

Figure 2.2.85 Symmetry elements of improper rotational point groups

Guide for the Determination of Point Groups

With the knowledge you have, you can unambiguously identify the point group of a molecule. The key to success is that you are able to see the symmetry elements in the molecule. This takes practice. With enough practice you can identify the point group of a molecule immediately. Until, you can use guides, that you can follow to identify a point group. Such a guide asks systematic questions about the presence or absence of a symmetry element. Depending on whether you answer the question with yes or no you can follow the guide in a particular direction. Eventually, after having answered enough questions the guide will lead you to the respective point group. You can see such a chart below (Fig. 2.2.86).

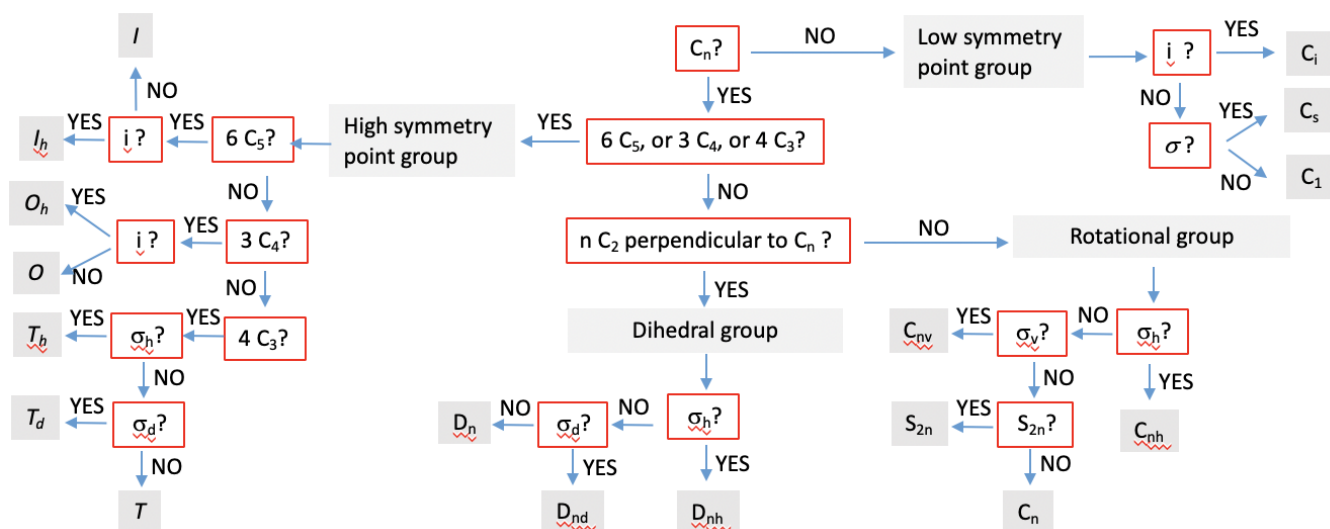


Figure 2.2.86 Guide for determination of point groups

You can first ask if there is at least one C_n present. If not, then the molecule must be in a low symmetry point group. If there is an inversion center, the point group is C_i . If not we can next ask, if there is a mirror plane. If yes it is C_s , and if not the point group is C_1 . If a low symmetry point group can be ruled out, then we can ask next, if there is a high symmetry point group. This is the case when there are either $4C_3$, $3C_4$, or $6C_5$ rotational axes present, standing for tetrahedral, octahedral, and icosahedral symmetry, respectively. If there is an inversion center in case a C_5 is present, the point group is I_h . If not, it is I . Similarly, if there are $3C_4$ axes, and an inversion center, the point group must be O_h . If there is no inversion center the point group is O . If there are $4C_3$ axes, the point group must be T type. If there is a horizontal mirror plane in addition, then the point group must be T_h . If not, we can ask next is there are dihedral mirror planes. If yes, the point group is T_d , otherwise it is T . Now we have checked for all high symmetry point groups.

If a high-symmetry point groups can be ruled out, we ask if there are $n C_2$ axes in addition to the C_n principal axis. If this is so, then we must have a dihedral group of the D type. Next we ask, if there is a horizontal mirror plane. If yes, the point group must be D_{nh} . If not, we ask if there are dihedral mirror planes. If yes, the point group is D_{nd} . If there are no mirror planes at all, then the point group is D_n . If there are no $n C_2$ axes in addition to the C_n , the the group must be either a rotational group or an improper rotation group. We next ask, if there is a horizontal mirror plane. If yes, then the point group is of the type C_{nh} . If not, we ask if there are vertical mirror planes. If yes, then the point group is C_{nv} . If that is not the case, we ask if there is an S_{2n} in addition to the C_n . If yes, the point group is S_{2n} . If not, it is C_n .

Example: Dibromonaphthalene

Let us practice the point group guide by one example. Let us look at the dibromonaphthalene molecule (Fig. 2.2.87).

What is the point group of

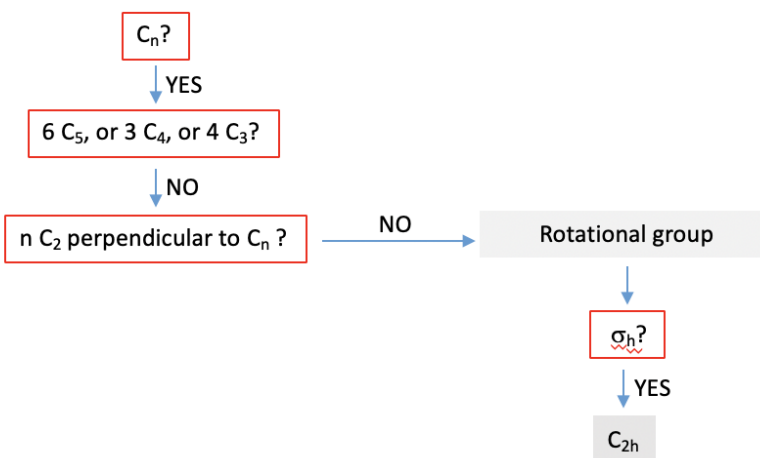
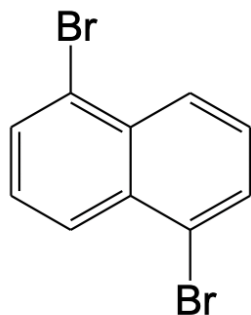


Figure 2.2.87 Guide for determining the point group of dibromonaphthalene

The first question we would ask is: Can you see at least one proper rotational axis? The answer is yes. There is a C_2 proper rotational axis that stands perpendicular to the plane of the molecule and goes through the center of the C-C bond that is shared by the two aromatic rings. Next we ask: Are the $6C_5$, $3C_4$ or $4C_3$ axes? That is clearly not the case, and thus we do not have a high symmetry point group. Next we can think about if there are $2C_2$ axes in addition to the C_2 axes we already found. The answer is no, so the point group cannot be a dihedral group. Next, we would ask: Is there a horizontal mirror plane. This is indeed the case. There is a horizontal mirror plane in the plane of the molecule. It does not move any atoms around, but as we discussed before, a mirror does not need to do this to exist. This identifies the point group as C_{2h} .

Chiral Point Groups

A chiral point group is a point group that only has proper rotation operations in addition to the identity. This is equivalent to the statement that no improper rotations must exist in a chiral point group. This includes mirror planes and inversion centers because a mirror plane is the same as an S_1 , and an inversion center is the same as an S_2 . If a molecule belongs to a chiral point group, then it has a mirror image that cannot be superimposed with the original molecule. The two mirror images are called enantiomers. An example is the bromochlorofluoromethane. You can see two enantiomers separated by a dotted line (Fig. 2.2.88).

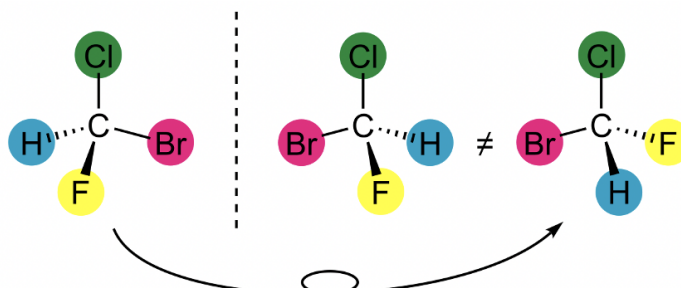


Figure 2.2.88 Chirality in bromochlorofluoromethane

The dotted line represents a mirror plane. Note that this mirror plane is not a mirror plane in the meaning of a symmetry element. You can see that the two molecules to the left and the right of the mirror plane are mirror images respective to each other. The molecules cannot be superimposed meaning that they are enantiomers. Note that the bromochlorofluoromethane molecule on the far right is not an enantiomer to the other two. It is the same molecule as the enantiomer on the far left. We only need to rotate clockwise around the C-Cl axis to make the two molecules superimpose meaning that they are the same.

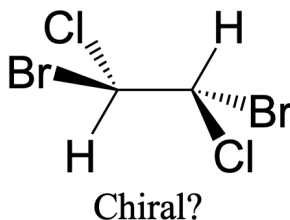


Figure 2.2.89 Dibromodichloroethane

It should be pointed out that the fact that a molecule has a carbon with four different substituents is not sufficient to make it a chiral molecule. The dibromodichloroethane molecule shown above (Fig. 2.2.89) has four different substituents around the carbon atom, but it is not chiral because there is an inversion center in the middle of the C-C bond.

Chiral High Symmetry Point Groups

Chiral groups not necessarily need to have low symmetry, in fact, the high symmetry rotational subgroups I, O, and T are chiral groups because they only have proper axes in addition to the identity (Fig. 2.2.90).

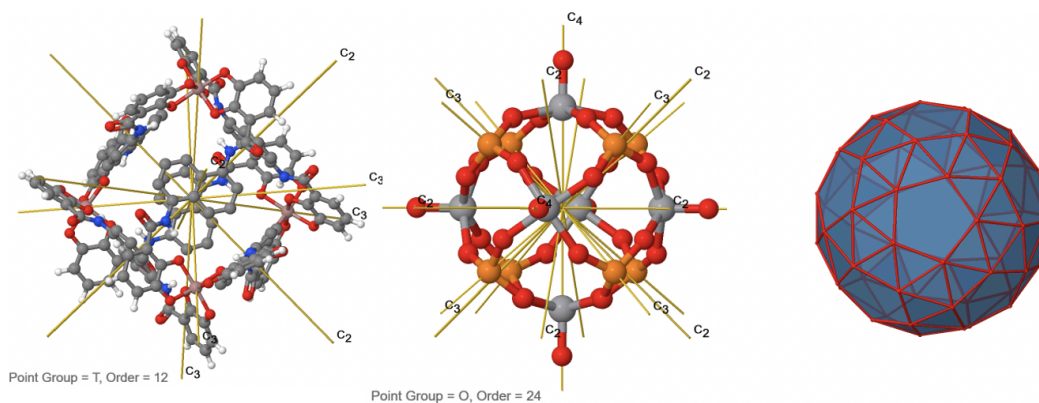


Figure 2.2.90 Objects belonging to chiral high symmetry point groups (T, O, and I from left to right).

Dr. Kai Landskron ([Lehigh University](#)). If you like this textbook, please consider to make a donation to support the author's research at Lehigh University: [Click Here to Donate](#).

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